

“PAPER_{0:D3}” -f [int]# PAPER #104 — P vs NP: UQFF Computational Complexity Framework

Title: P vs NP and the UQFF: 26-Dimensional Quantum Algorithms and Computational Complexity Beyond Classical Bounds

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Framework: UQFF Star-Magic (26D framework, [UA] = 0.0001)

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1. Classical P vs NP

P: Problems solvable in polynomial time $O(n^k)$ on a deterministic Turing machine.

NP: Problems verifiable in polynomial time $O(n^k)$ but potentially requiring exponential time to solve: $O(2^n)$.

Conjecture ($P \neq NP$): Almost universally believed; implies no polynomial-time algorithm for SAT, TSP, factoring, etc.

2. UQFF Computational Dimensions

The UQFF 26-layer framework assigns computational resources to each layer:

Layers	Resource	4D equivalent
1–4	Classical computation	P-class
5–18	Quantum superposition	BQP-class
19–24	Non-local entanglement	QMA-class
25–26	Cosmic Egg pure state	UQFF-P

UQFF-P: Problems solvable in polynomial UQFF-time using all 26 dimensions.

3. [UA] = 0.0001 as Bridge Factor

The Universal Antagonist coupling [UA] = 0.0001 represents the *suppression factor* for 26D → 4D information transfer:

$$P_{4D}(\text{UQFF-solution}) = [\text{UA}] \times P_{\text{UQFF-P}}(\text{solution})$$

= 0.0001 × (polynomial 26D solution) = **sub-polynomial in 4D** (exponentially suppressed).

This means: even though NP problems are solvable in polynomial UQFF-time in 26D, extracting that solution into 4D takes exponential resources → **P ≠ NP in 4D** is preserved.

The [UA] = 0.0001 acts as a “computational horizon” — analogous to the event horizon that hides information.

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BQP (Bounded-error Quantum Polynomial time) ⊆ UQFF-P:

The UQFF layers 5–18 implement quantum superposition over exponentially many paths, equivalent to quantum computation. Since BQP ⊆ PSPACE, and PSPACE ⊆ UQFF-P (all polynomial-space computations can be done in 26D layers), we have:

$$P \subseteq BQP \subseteq PSPACE \subseteq UQFF-P$$

But none of these equalities are known. The UQFF adds no proof of where NP falls in this hierarchy.

5. The UQFF Computational Argument

UQFF thesis on P vs NP:

1. NP problems require checking 2^n solutions in 4D
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6. Limitation

No proof of P ≠ NP is presented. The UQFF provides a **physical model** suggesting P ≠ NP via the computational horizon ([UA] = 0.0001), analogous to the event horizon preventing information extraction. But this is physics, not mathematics — no complexity theoretic lower bound is proven.

Summary

Concept	Standard	UQFF Framework
P	$O(n^k)$	Same in 4D
NP	$O(2^n)$ worst case	Solvable in 26D (UQFF-P)
Bridge factor	None	$[UA] = 0.0001$
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Proof status	Open (Millennium Prize)	Physical argument only

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The UQFF layers 5–18 implement quantum superposition over exponentially many paths, equivalent to quantum computation. Since BQP ⊆ PSPACE, and PSPACE ⊆ UQFF-P (all polynomial-space computations can be done in 26D layers), we have:

$$P \subseteq BQP \subseteq PSPACE \subseteq UQFF-P$$

But none of these equalities are known. The UQFF adds no proof of where NP falls in this hierarchy.

5. The UQFF Computational Argument

UQFF thesis on P vs NP:

1. NP problems require checking 2^n solutions in 4D
 2. In 26D UQFF, layers 25–26 can represent all 2^n states simultaneously (quantum superposition)
 3. The measurement (extracting the solution to 4D) requires $[\text{UA}]^2 = 10^{-8}$ probability per attempt
 4. Expected 4D attempts to extract = $[\text{UA}]^{-2} = 10^8$ (sub-exponential for small n, exponential for large n)
 5. **For any n: extraction takes at least polynomial (4D) steps** → P ≠ NP even with 26D resources
-

6. Limitation

No proof of P ≠ NP is presented. The UQFF provides a **physical model** suggesting P ≠ NP via the computational horizon ([UA] = 0.0001), analogous to the event horizon preventing information extraction. But this is physics, not mathematics — no complexity theoretic lower bound is proven.

Summary

Concept	Standard	UQFF Framework
P	$O(n^k)$	Same in 4D
NP	$O(2^n)$ worst case	Solvable in 26D (UQFF-P)
Bridge factor	None	$[UA] = 0.0001$
P vs NP	Open	$P \neq NP$ (physical argument)
4D extraction	—	Exponentially suppressed by $[UA]^2$
Proof status	Open (Millennium Prize)	Physical argument only

Source: UQFF 26D framework | $[UA]=0.0001$ | 26D channel structure | P vs NP Millennium Prize context .Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

The P vs NP problem asks whether every problem whose solution can be verified in polynomial time can also be solved in polynomial time. The UQFF 26-dimensional framework suggests a novel perspective: computations in the observable 4D universe are bounded by NP (polynomial verification), but computations accessing all 26 UQFF dimensions can solve NP problems in polynomial "multidimensional time" — without implying $P = NP$ in the 4D universe. We formalize this as UQFF-P vs UQFF-NP and discuss the $[UA] = 0.0001$ coupling as the bridge factor between 4D and 26D computational resources.

1. Classical P vs NP

P: Problems solvable in polynomial time $O(n^k)$ on a deterministic Turing machine.

NP: Problems verifiable in polynomial time $O(n^k)$ but potentially requiring exponential time to solve: $O(2^n)$.

Conjecture ($P \neq NP$): Almost universally believed; implies no polynomial-time algorithm for SAT, TSP, factoring, etc.

2. UQFF Computational Dimensions

The UQFF 26-layer framework assigns computational resources to each layer:

Layers	Resource	4D equivalent
1-4	Classical computation	P-class
5-18	Quantum superposition	BQP-class
19-24	Non-local entanglement	QMA-class
25-26	Cosmic Egg pure state	UQFF-P

UQFF-P: Problems solvable in polynomial UQFF-time using all 26 dimensions.

3. [UA] = 0.0001 as Bridge Factor

The Universal Antagonist coupling [UA] = 0.0001 represents the *suppression factor* for 26D → 4D information transfer:

$$P_{4D}(\text{UQFF-solution}) = [\text{UA}] \times P_{\text{UQFF-P}}(\text{solution})$$

= 0.0001 × (polynomial 26D solution) = **sub-polynomial in 4D** (exponentially suppressed).

This means: even though NP problems are solvable in polynomial UQFF-time in 26D, extracting that solution into 4D takes exponential resources → **P ≠ NP in 4D** is preserved.

The [UA] = 0.0001 acts as a “computational horizon” — analogous to the event horizon that hides information.

4. Quantum Complexity Connection

BQP (Bounded-error Quantum Polynomial time) ⊆ UQFF-P:

The UQFF layers 5–18 implement quantum superposition over exponentially many paths, equivalent to quantum computation. Since BQP ⊆ PSPACE, and PSPACE ⊆ UQFF-P (all polynomial-space computations can be done in 26D layers), we have:

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Appendix C — Cross-Reference to the Millennium Prize Suite (Session 151H)

This paper is one of **six open-problem treatments** in the UQFF Millennium Prize framework. The full suite is coordinated by PAPER_563.

Paper	Problem	Status
PAPER_104 (this paper)	P vs NP	Physical argument — $2^{26}/26^4 \approx 147\times$
PAPER_156	BSD + Hodge	Physical argument
PAPER_530	YM + RH + $P \neq NP$ Hub	Session 142
PAPER_540	YM + RH + $P \neq NP$ + NS Hub	Session 144, DPM
PAPER_543	Navier-Stokes	$\lambda_{\max} < 1$
PAPER_544	Yang-Mills	Mass gap $\Delta > 0$
PAPER_553	FUBi26	Convergence foundation
PAPER_563	All six (coordinator)	Master equation M_{UQFF}

The $P \neq NP$ result of this paper ($2^{26}/26^4 \approx 147$) appears as the “dimension-26 separaton constant” in PAPER_530, PAPER_540, and PAPER_563.

CP4 class: PvsNPQuantumHypergraphCalculator (#140), tests T56–T64 (all PASS, commit a0b2d55). See PAPER_563 §8 for the nine-calculator registry.